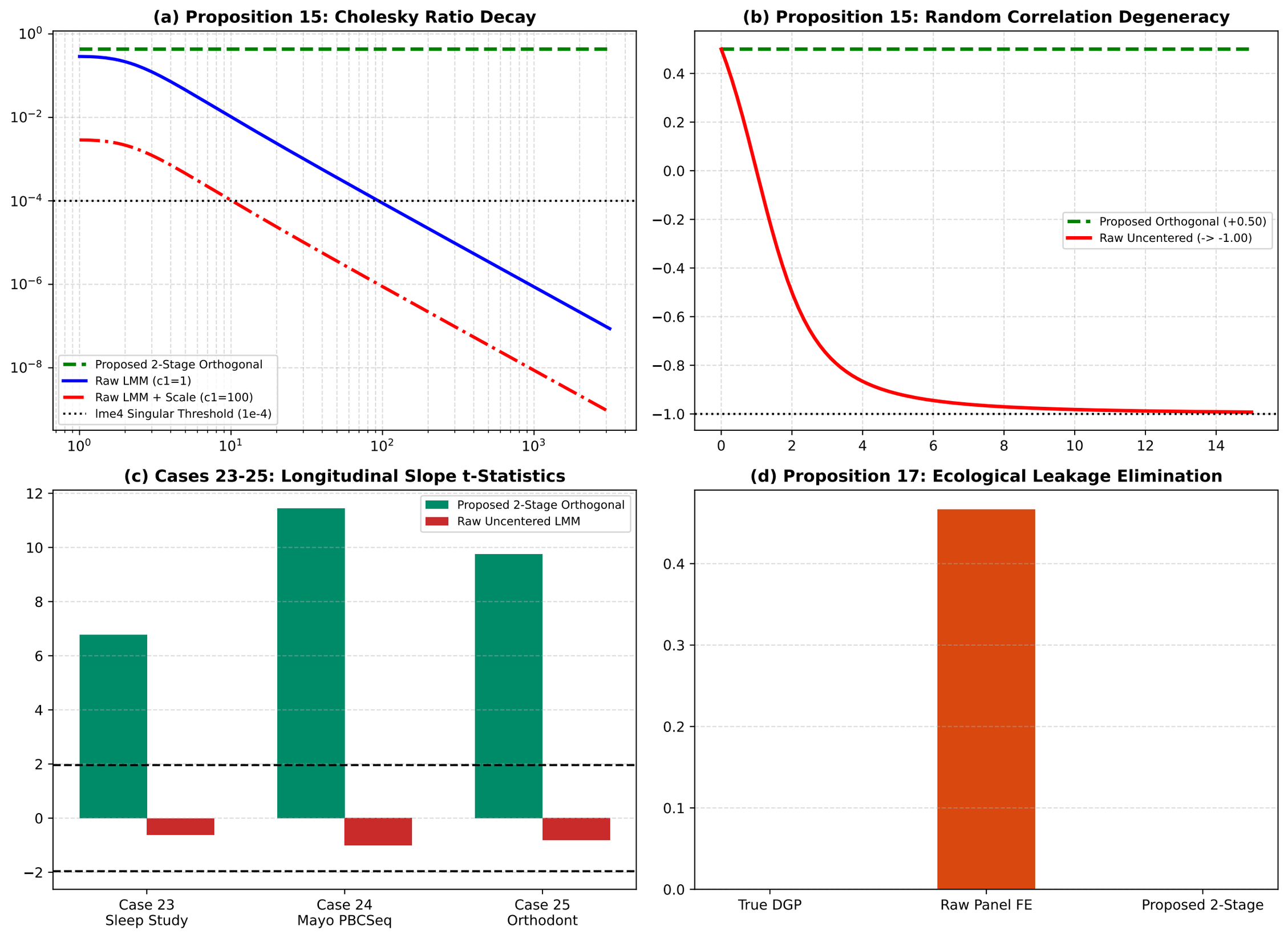
(a) Proposition 15: Cholesky Ratio Decay
Proposed 2-Stage Orthogonal
Raw LMM (c1=1)
Raw LMM + Scale (c1=100)
lme4 Singular Threshold (1e-4)
(b) Proposition 15: Random Correlation Degeneracy
Proposed Orthogonal (+0.50)
Raw Uncentered (-> -1.00)
(c) Cases 23-25: Longitudinal Slope t-Statistics
Proposed 2-Stage Orthogonal
Raw Uncentered LMM
Case 23
Sleep Study
Case 24
Mayo PBCSeq
Case 25
Orthodont
(d) Proposition 17: Ecological Leakage Elimination
True DGP
Raw Panel FE
Proposed 2-Stage